Kubernetes Troubleshooting Playbook

11 production incidents — symptom → cause → fix

SRE Investigation Workflow · Enterprise Edition

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Audience: Beginner → Engineer → Enterprise
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Node NotReady

Symptom. kubelet stopped reporting; pods on the node enter Unknown then Terminating.

Likely causes. 1. Disk pressure 2. kubelet crash 3. CNI plugin failure 4. Network partition

Investigation commands:

```
ssh node && journalctl -u kubelet -n 200
systemctl status kubelet containerd
df -h /var/lib/kubelet
```

CrashLoopBackOff

Symptom. Pod restarts repeatedly with exponential back-off (10s → 20s → 40s …).

Likely causes. Bad command, missing env/secret, failing liveness probe, app exception

Investigation commands:

```
kubectl logs <pod> --previous
kubectl describe pod <pod> | grep -A5 'Last State'
kubectl get events --field-selector involvedObject.name=<pod>
```

ImagePullBackOff

Symptom. Pod stays in ContainerCreating with image pull errors.

Likely causes. Wrong tag, private registry without imagePullSecret, network egress blocked

Investigation commands:

```
kubectl describe pod <pod>
kubectl create secret docker-registry regcred --docker-server=...
kubectl patch sa default -p '{"imagePullSecrets":[{"name":"regcred"}]}'
```

Pending Pods

Symptom. Scheduler can't place the pod on any node.

Likely causes. Insufficient CPU/memory, taints without tolerations, PVC not bound, nodeSelector mismatch

Investigation commands:

```
kubectl describe pod <pod> | tail -20
kubectl get nodes -o json | jq '.items[].status.allocatable'
```

CoreDNS Failure

Symptom. Pods can't resolve cluster DNS; service-to-service calls time out.

Likely causes. CoreDNS pods crashed, forwarder upstream down, NetworkPolicy blocking 53

Investigation commands:

```
kubectl -n kube-system logs deploy/coredns
kubectl exec -it dnsutils -- dig kubernetes.default
```

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CNI Failure

Symptom. New pods stuck in ContainerCreating; existing pods lose connectivity.

Likely causes. CNI binary missing, plugin crash, IP pool exhausted

Investigation commands:

```
ls /opt/cni/bin
journalctl -u kubelet | grep -i cni
kubectl -n kube-system get pods | grep -E 'cilium|calico|flannel'
```

OOMKilled

Symptom. Container killed by Linux OOM; exit code 137.

Likely causes. memory limit too low, memory leak, JVM heap exceeds container limit

Investigation commands:

```
kubectl describe pod <pod> | grep -A3 'Last State'
kubectl top pod <pod>
```

Probe Failures

Symptom. Pod becomes Unready or restarts; endpoints drop traffic.

Likely causes. Wrong path/port, slow startup, initialDelaySeconds too low

Investigation commands:

```
kubectl describe pod <pod> | grep -A2 Liveness
kubectl logs <pod> | tail -50
```

PVC Pending

Symptom. Volume never binds; pod stays Pending.

Likely causes. No StorageClass, CSI driver crashed, access mode mismatch, cloud quota

Investigation commands:

```
kubectl get sc
kubectl describe pvc <name>
kubectl -n kube-system logs ds/<csi-node>
```

API Server 5xx

Symptom. kubectl returns Internal error; controllers stop reconciling.

Likely causes. etcd unhealthy, audit log disk full, webhook timeout

Investigation commands:

```
kubectl -n kube-system get pods
etcdctl endpoint health
```

```
journalctl -u kube-apiserver -n 200
```

Service Unreachable

Symptom. Service IP times out even though pods are Running.

Likely causes. Selector mismatch (no endpoints), kube-proxy down, NetworkPolicy deny

Investigation commands:

```
kubectl get endpoints <svc>
kubectl -n kube-system get pods | grep kube-proxy
kubectl get networkpolicy -A
```